To Fume or Not to Fume
COLORING WITH STAINS

TEXT AND PHOTOS BY MAC SIMMONS



The top half of the upper panel is fumed with 26% ammonia for 5 hours; the lower half of that panel for 10 hours. The bottom panel shows the further darkening from a top coat of tung oil.

Fuming is a chemical coloring technique in which the gas fumes from liquid ammonia react with the natural tannin in certain woods to give it a distinctive color. This technique was especially popular for finishing oak furniture of the Craftsman era, and the term "fumed oak" is often used to describe this color in general. However, the major disadvantage of this method is that fuming requires strong chemicals and lots of labor. In this article, I will discuss an alternative method for simulating some of those fumed colors by using pigmented paste colorants to make your own easy-to-use wiping stains.

THE BASICS OF FUMING

If you are not familiar with how a true fumed finish is produced, here is a brief outline of the procedure. A clear plastic tent with a timber frame is erected in an outdoor location to house the workpiece during the fuming process. Household ammonia can be used, but it is a very weak solution that only contains about 5-10% ammonia. Professional finishers use an "industrial strength" 26-30% ammonia solution, which can be obtained at some office supply or blueprint supply stores. The work is first placed inside the clear plastic tent, and then small plastic or glass containers with covers are spaced evenly around the inside the tent. Ammonia is then added to the containers. You will need to construct the tent with a door or a flap which can be opened and closed so that you can reach inside the tent when you fill or refill these containers, and then be able to seal up the tent to keep the ammonia fumes inside.

You will also need to regularly check the material being fumed for color; the longer the wood remains inside the tent and the more tannin present in the wood, the darker the wood will become. The fumed oak wood samples at left show a panel where the top half was placed in the fuming tent for 5 hours with 26% ammonia, while the bottom half was fumed for 10 hours with the same ammonia; the effects of length of time and the amount of tannin in the wood is clearly visible. These two samples were not top-coated in order to show the natural fuming color. In the lower panel, however, you can see how the fuming appears after it is protected with a clear coat, in this case tung oil. Notice how much the panel further darkens after the top coating.

A few more points should be mentioned. Make sure there is no metal hardware on your workpieces when you fume them—the ammonia can create a black stain on the wood surrounding the metal. Whenever you're checking the woods for color, or when you are not fuming, use the lids to cover the containers. Remember, ammonia is a potent chemical which can be toxic, so you want to avoid exposure to your lungs, eyes, and skin. Always wear protective gear—respirator, goggles, and rubber gloves—whenever you are handling ammonia. This is not a project to be taken lightly; if you do intend to try fuming your wood, I strongly suggest that you do some further research into the process.

SIMULATING FUMED COLORS WITH STAINS

Now, its time to learn how you can use pigmented paste colorants to make up your own wiping stains which will do a good job of simulating the look of fumed wood.

The color of fumed oak can range from a tan to a dark walnut, which, as we have seen, will further darken when it is clear-coated. Stains, on the other hand, will usually look the same color (perhaps a shade or two darker) after they are clear-coated.

For compatibility with solvent-based top coatings, you can use any one of the paste-pigmented colorants, such as Universal Tinting Colors (UTC), oil or Japan colors. For use with water-borne top coatings, either acrylic or artist water colorants will work. Then select the appropriate compatible solvent to add to your colorant to make up your own wiping stains. A basic formula for a wiping stain uses approximately one part of colorant (Universal, Oil, Japan) to six parts of mineral spirits.

When working with stains, there are a few general points to keep in mind. Since all colorants are not the same in shade or in strength, you will have to adjust them individually, adding more solvent to dilute the color if it is too dark, or more colorant if it is too light. Always work in small increments until the targeted color is reached. Mix well, and keep the container tightly covered when not in use. Always record the formula in case it is needed again, and always mix up enough stain at one time to complete your project. I strongly recommend that you always make up complete start-to-finish samples, so that you can see the final color after top-coating. And finally, be sure you always wear proper protective gear whenever you're finishing.

Since the fuming process naturally produces dissimilar colors according to the amount of tannic acid in different portions of the wood, you may require a variety of wiping stains if you are trying to do exact color matching of a fumed piece. The photograph at the top right shows three different colored stains that were applied on quartersawn oak, which were stained and then clear-coated with lacquer.

To match a dark fumed oak sample, I use a Van Dyke Brown and a black colorant; I begin with the Van Dyke Brown colorant and then reduce the color strength by 50% with a mineral spirit solvent to make up the lighter tan color. I then add an ounce of the black colorants to the quart of Van Dyke Brown stain to make up the darker brown color in the stained colored sample.

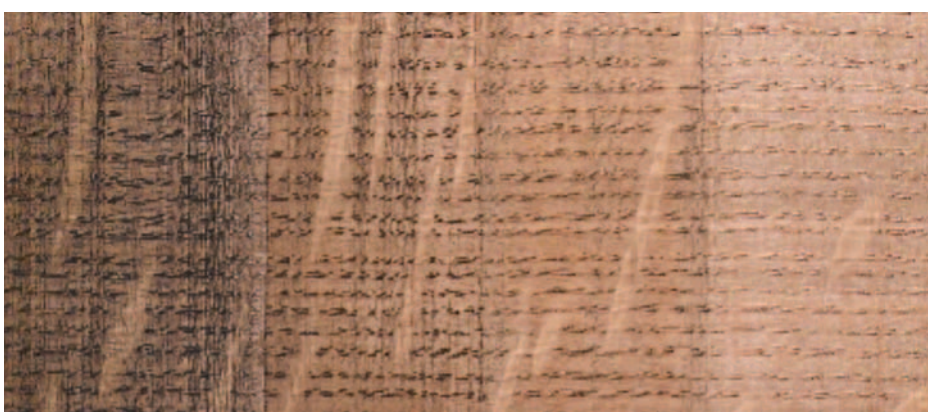
The middle sample has a Burnt Umber stain, and the bottom sample is a straight Van Dyke Brown stain. For all three samples, apply the stain and then wipe off the excess; wiping it dry will even out the color and allow the stain to dry faster, so that it can be top-coated in a few hours.

In cases where you need to get more of a tan color, you can add a little of the Van Dyke Brown into a white colorant and mix it thoroughly, which will give you many shades of tan. Adding a little French Yellow Ochre or Orange to that same tan mix will give you many additional shades. You will find that simply by diluting the colored stains with additional solvent, you will be able to match many color samples. Again, make up different color samples and keep careful record of the formulation for future use

Top coatings also play an important part in achieving the final color. Clear or orange shellac was commonly used over fumed woods, which contributed to its distinctive color. Oil finishes such as Tung, Boiled Linseed, and the Danish Oils will add coatings with different color. You can even add different colorants to clear oil finishes to produce subtle changes to complete your matching of fumed finishes.



AT LEFT—Three wiping stains for simulating fumed oak: Van Dyke Brown plus Black (top), Burnt Umber (middle), and Van Dyke Brown (bottom). All are top-coated with lacquer.



BELOW—Simply diluting a wiping stain produces many more shades of color.

THERE IS A DIFFERENCE

I must admit that fuming is a unique coloring technique that has its own place in finishing. But for those finishers who prefer not to use the fuming process, I think you will find that simulating any of the fumed colors by staining is not a difficult task to learn. I found it fairly easy to do; as always, experimentation and keeping careful records and samples are the key. I know that you also will find it a good alternative once you give it a try.

Mac Simmons is a 45-year veteran of the furniture finishing, refinishing, and restoration trades, and writes for several magazines. His book, Fearless Finishing, *is scheduled for publication by F&W Publishing in 2006.*